

Chapter 1: Artificial Intelligence & Agentic Workflows

1. Overview of Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines programmed to think, learn, and solve problems. Traditional AI models focus on single-task processing, such as text generation, image recognition, or rule-based decision logic.

Core Subfields of AI:

- **Machine Learning (ML):** Algorithms that improve automatically through experience and data analysis.
- **Natural Language Processing (NLP):** Systems enabling computers to understand, interpret, and manipulate human language.
- **Computer Vision:** Techniques that allow software to process, analyze, and extract meaning from visual inputs.

2. Agentic AI and Autonomous Systems

Agentic AI represents a paradigm shift from passive response models to proactive execution systems. Unlike standard conversational bots, agentic systems maintain autonomy to complete multi-step goals without direct human intervention at every step.

Key Capabilities of NOVA:

- **Tool Augmentation:** Dynamically invokes external functions such as Web Search, YouTube search, Wikipedia retrieval, local File Management, long-term Memory, and RAG document stores.
- **Retrieval-Augmented Generation (RAG):** Integrates vector databases (such as ChromaDB) to query external documents—including PDFs, PowerPoint presentations, and text files—to ground responses in factual study material.